

Coupled Cable Equations in the Witten–Rozansky Framework: an LLM Study

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Abstract

In this note, the authors consider N coupled cables with arbitrary inclinations in Euclidean 3-space. Mimicking some of the work in [1], the authors guide an LLM to generate the N final coupled equations.

1 Introduction and Motivation

In the framework inspired by the Witten–Rozansky paper [2], we study the dynamics of voltage propagation along one-dimensional structures (“cables”) embedded in an ambient three-dimensional space, $\mathbb{R}^3$. In our model the cables are represented as tilted cylinders with intrinsic coordinates. An embedding of a single cable is given by

$$\phi(s) = (x(s), y(s), z(s)).$$

By the chain rule the derivative along the cable is

$$\frac{d}{ds} = \frac{dx}{ds} \frac{\partial}{\partial x} + \frac{dy}{ds} \frac{\partial}{\partial y} + \frac{dz}{ds} \frac{\partial}{\partial z}.$$

This idea parallels the Rozansky–Witten sigma model in which the target geometry is pulled back via a map $\phi : M \rightarrow X$. (In our context one may also view the target as a submanifold of $\mathbb{R}^3$.)

2 The Cable Equation on a Single (Tilted) Cylinder

The standard cable equation (with time τ) in one spatial dimension is

$$C_m \frac{\partial V}{\partial \tau} = \frac{a}{2R_i} \frac{\partial^2 V}{\partial s^2} - \frac{V}{R_m},$$

where V is the voltage, C_m the membrane capacitance, and R_i , R_m are the axial and membrane resistances, respectively.

When the cable is tilted in $\mathbb{R}^3$, we define the unit tangent vector

$$T = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta),$$

so that

$$\frac{\partial}{\partial s} = T \cdot \nabla, \quad \frac{\partial^2}{\partial s^2} = (T \cdot \nabla)^2.$$

3 Coupling Two Cables

For two distinct cables, Cylinder 1 (with coordinate s) and Cylinder 2 (with coordinate t), each obeys its own cable equation:

$$C_m \frac{\partial V_1}{\partial \tau} = \frac{a}{2R_i} \frac{\partial^2 V_1}{\partial s^2} - \frac{V_1}{R_m},$$

$$C_m \frac{\partial V_2}{\partial \tau} = \frac{a}{2R_i} \frac{\partial^2 V_2}{\partial t^2} - \frac{V_2}{R_m}.$$

We cross-couple the two by adding a fraction κ of the other cable's second derivative. That is, in the equation for V_1 we add $\kappa \frac{\partial^2 V_2}{\partial t^2}$ and vice versa.

4 Change of Coordinates: Mean and Difference

Define the *mean* and *difference* coordinates by

$$m = \frac{s+t}{2}, \quad \Delta = \frac{s-t}{2},$$

so that

$$s = m + \Delta, \quad t = m - \Delta.$$

Then by the chain rule,

$$\frac{\partial}{\partial s} = \frac{1}{2} \frac{\partial}{\partial m} + \frac{1}{2} \frac{\partial}{\partial \Delta}, \quad \frac{\partial}{\partial t} = \frac{1}{2} \frac{\partial}{\partial m} - \frac{1}{2} \frac{\partial}{\partial \Delta}.$$

Squaring gives:

$$\frac{\partial^2}{\partial s^2} = \frac{1}{4} \frac{\partial^2}{\partial m^2} + \frac{1}{2} \frac{\partial^2}{\partial m \partial \Delta} + \frac{1}{4} \frac{\partial^2}{\partial \Delta^2},$$

$$\frac{\partial^2}{\partial t^2} = \frac{1}{4} \frac{\partial^2}{\partial m^2} - \frac{1}{2} \frac{\partial^2}{\partial m \partial \Delta} + \frac{1}{4} \frac{\partial^2}{\partial \Delta^2}.$$

Thus, the coupled cable equations for the two cylinders become:

Cylinder 1:

$$C_m \frac{\partial V_1(m, \Delta, \tau)}{\partial \tau} = \frac{a}{2R_i} \left\{ \frac{1}{4} \frac{\partial^2 V_1}{\partial m^2} + \frac{1}{2} \frac{\partial^2 V_1}{\partial m \partial \Delta} + \frac{1}{4} \frac{\partial^2 V_1}{\partial \Delta^2} \right. \\ \left. + \kappa \left[\frac{1}{4} \frac{\partial^2 V_2}{\partial m^2} - \frac{1}{2} \frac{\partial^2 V_2}{\partial m \partial \Delta} + \frac{1}{4} \frac{\partial^2 V_2}{\partial \Delta^2} \right] \right\} - \frac{V_1(m, \Delta, \tau)}{R_m}.$$

Cylinder 2:

$$C_m \frac{\partial V_2(m, \Delta, \tau)}{\partial \tau} = \frac{a}{2R_i} \left\{ \frac{1}{4} \frac{\partial^2 V_2}{\partial m^2} - \frac{1}{2} \frac{\partial^2 V_2}{\partial m \partial \Delta} + \frac{1}{4} \frac{\partial^2 V_2}{\partial \Delta^2} \right. \\ \left. + \kappa \left[\frac{1}{4} \frac{\partial^2 V_1}{\partial m^2} + \frac{1}{2} \frac{\partial^2 V_1}{\partial m \partial \Delta} + \frac{1}{4} \frac{\partial^2 V_1}{\partial \Delta^2} \right] \right\} - \frac{V_2(m, \Delta, \tau)}{R_m}.$$

5 Generalization to N Cables

For N cables with intrinsic coordinates $s_1, s_2, \dots, s_N$, define the mean coordinate

$$m = \frac{1}{N} \sum_{i=1}^N s_i,$$

and deviations

$$\delta_i = s_i - m, \quad i = 1, \dots, N, \quad \text{with} \quad \sum_{i=1}^N \delta_i = 0.$$

Since there are $N-1$ independent deviations, one can introduce an $(N-1)$ -dimensional vector

$$\Delta = (\Delta_1, \Delta_2, \dots, \Delta_{N-1}),$$

so that

$$s_i = m + \sum_{j=1}^{N-1} A_{ij} \Delta_j,$$

where the matrix A_{ij} satisfies $\sum_{i=1}^N A_{ij} = 0$. Then, by the chain rule,

$$\frac{\partial}{\partial s_i} = \frac{1}{N} \frac{\partial}{\partial m} + \sum_{j=1}^{N-1} A_{ij} \frac{\partial}{\partial \Delta_j}.$$

Hence,

$$\frac{\partial^2}{\partial s_i^2} = \frac{1}{N^2} \frac{\partial^2}{\partial m^2} + \frac{2}{N} \sum_{j=1}^{N-1} A_{ij} \frac{\partial^2}{\partial m \partial \Delta_j} + \sum_{j,k=1}^{N-1} A_{ij} A_{ik} \frac{\partial^2}{\partial \Delta_j \partial \Delta_k}.$$

If, in the uncoupled case, each cable satisfies

$$C_m \frac{\partial V_i}{\partial \tau} = \frac{a}{2R_i} \frac{\partial^2 V_i}{\partial s_i^2} - \frac{V_i}{R_m},$$

and we introduce cross-coupling by adding a term κ times the second derivative from every other cable, then the coupled equation for cable i is

$$\begin{aligned} C_m \frac{\partial V_i(m, \Delta, \tau)}{\partial \tau} = & \frac{a}{2R_i} \left\{ \frac{1}{N^2} \frac{\partial^2 V_i}{\partial m^2} + \frac{2}{N} \sum_{j=1}^{N-1} A_{ij} \frac{\partial^2 V_i}{\partial m \partial \Delta_j} + \sum_{j,k=1}^{N-1} A_{ij} A_{ik} \frac{\partial^2 V_i}{\partial \Delta_j \partial \Delta_k} \right. \\ & + \kappa \sum_{l \neq i} \left[\frac{1}{N^2} \frac{\partial^2 V_l}{\partial m^2} + \frac{2}{N} \sum_{j=1}^{N-1} A_{lj} \frac{\partial^2 V_l}{\partial m \partial \Delta_j} + \sum_{j,k=1}^{N-1} A_{lj} A_{lk} \frac{\partial^2 V_l}{\partial \Delta_j \partial \Delta_k} \right] \Big\} \\ & - \frac{V_i(m, \Delta, \tau)}{R_m}. \end{aligned}$$

6 Re-expression in Terms of Ambient Coordinates (Angles Visible)

In our original setting, each cable is embedded in $\mathbb{R}^3$ by

$$\phi_i(s_i) = (x_i(s_i), y_i(s_i), z_i(s_i)).$$

Then by the chain rule

$$\frac{\partial}{\partial s_i} = \frac{dx_i}{ds_i} \frac{\partial}{\partial x} + \frac{dy_i}{ds_i} \frac{\partial}{\partial y} + \frac{dz_i}{ds_i} \frac{\partial}{\partial z}.$$

Defining the unit tangent vector

$$T_i = (\sin \theta_i \cos \phi_i, \sin \theta_i \sin \phi_i, \cos \theta_i),$$

we have

$$\frac{\partial}{\partial s_i} = T_i \cdot \nabla, \quad \frac{\partial^2}{\partial s_i^2} = (T_i \cdot \nabla)^2.$$

If we now introduce cross-coupling by adding to cable i a term κ times the intrinsic second derivative from every other cable, the final coupled cable equation for cable i (in ambient Cartesian coordinates) is

$$C_m \frac{\partial V_i}{\partial \tau} = \frac{a}{2R_i} \left[(T_i \cdot \nabla)^2 V_i + \kappa \sum_{j \neq i} (T_j \cdot \nabla)^2 V_j \right] - \frac{V_i}{R_m}.$$

Writing the tangent operators explicitly in terms of the angles, we obtain

$$C_m \frac{\partial V_i}{\partial \tau} = \frac{a}{2R_i} \left[(\sin \theta_i \cos \phi_i \partial_x + \sin \theta_i \sin \phi_i \partial_y + \cos \theta_i \partial_z)^2 V_i + \dots \right] \\ \left[\dots \kappa \sum_{j \neq i} (\sin \theta_j \cos \phi_j \partial_x + \sin \theta_j \sin \phi_j \partial_y + \cos \theta_j \partial_z)^2 V_j \right] - \frac{V_i}{R_m}.$$

7 Conclusion

The final equation above describes the dynamics of N coupled cables in $\mathbb{R}^3$. Each cable diffuses according to its own intrinsic orientation (explicitly given by the angles θ_i and ϕ_i) and is cross-coupled to all other cables through the term weighted by κ . This provides a unified description in ambient coordinates where the geometric orientation of each cable is visible.

References

- [1] Aman Chawla, Salvatore Morgera, and Arthur Snider. On axon interaction and its role in neurological networks. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2019.
- [2] Lev Rozansky and Edward Witten. Hyper-kähler geometry and invariants of three-manifolds. *Selecta Mathematica*, 3:401–458, 1997.